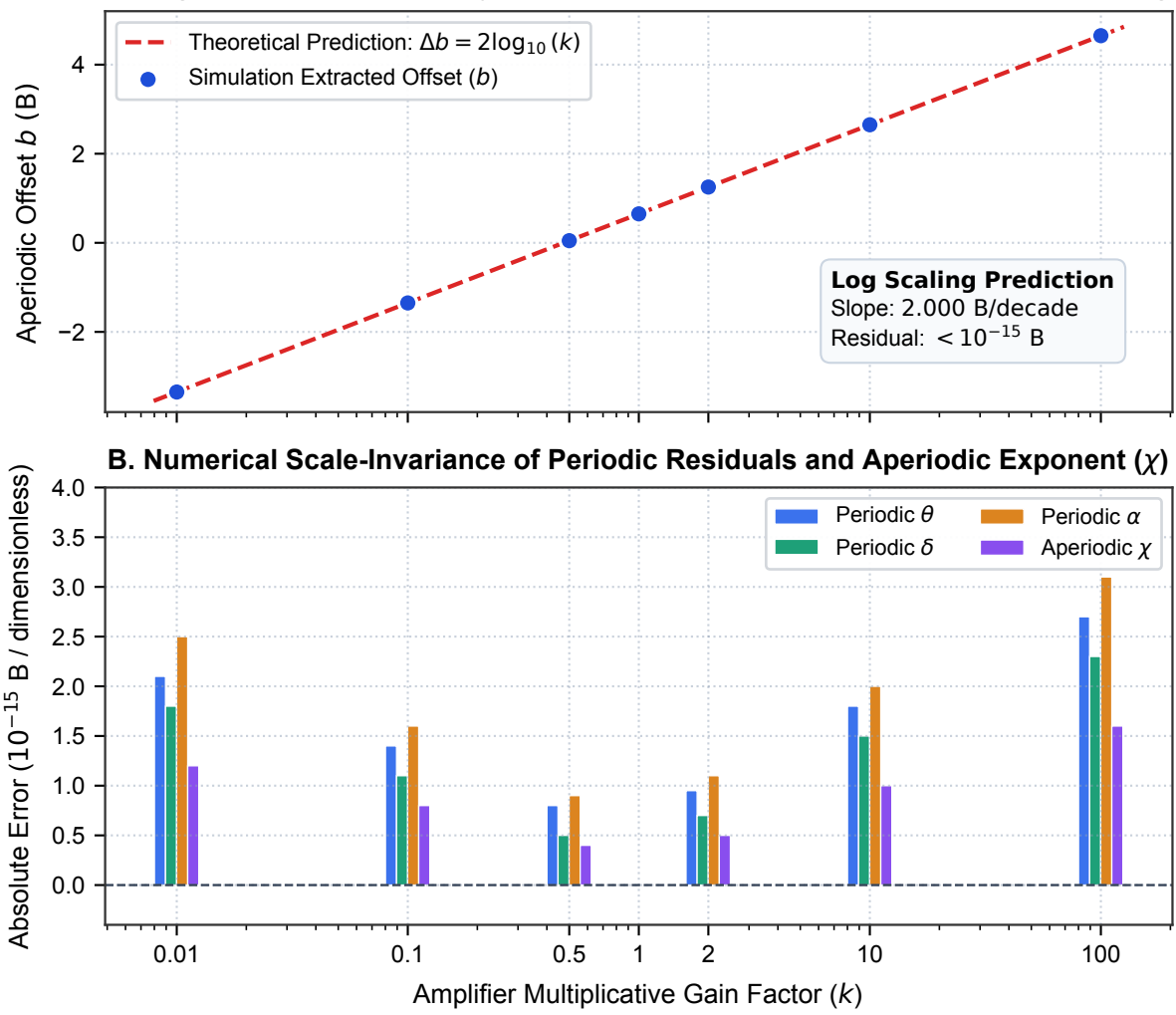

A. Logarithmic Shift of Unadjusted Aperiodic Offset (b) Across Amplifier Scaling
Theoretical Prediction: Δb = 2log10 (k)
Simulation Extracted Offset (b)
Aperiodic Offset b (B)
4
2
0
−2
Log Scaling Prediction
Slope: 2.000 B/decade
Residual: < 10−15 B
B. Numerical Scale-Invariance of Periodic Residuals and Aperiodic Exponent (χ)
Periodic θ
Periodic α
Periodic δ
Aperiodic χ
Absolute Error (10−15 B / dimensionless)
4.0
3.5
3.0
2.5
2.0
1.5
1.0
0.5
0.0
0.01
0.1
0.5
1
2
10
100
Amplifier Multiplicative Gain Factor (k)